Roy Huang

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Education

University of Michigan, College of Engineering **Ann Arbor, MI**

- B.S.E., Computer Science, GPA: 3.87, University Honors Apr 2026
- Coursework: Advanced Operating Systems, Distributed Systems, Cybersecurity, Formal Verification

Work Experience

MathWorks **Natick, MA**

Software Engineer Intern May 2025 - Aug 2025

- Designed and implemented a C/C++ cryptographic security component, enhancing code signing and verification.
- Reduced code signing component time by 20% through performance benchmarking and library optimizations.
- Utilized cryptographic signature tools to analyze MathWorks binary code signing on macOS and Linux.
- Developed a Simulink code generation tool to compile Simulink models to C++, JavaScript, and WebAssembly.

Conjuguemos **Newton, MA**

Software Engineer Intern June 2024 - Aug 2024

- Designed 5 new educational games, created online demos (HTML/JS), and presented work to senior stakeholders.
- Led team project to successfully obtain SOC2 security compliance and meet online accessibility requirements.
- Provided development and user-oriented feedback to enhance user experience and learning through gamification.

KTBYTE Computer Science Academy **Lexington, MA**

Programming Teacher Oct 2021 - Present

- Led weekly office hours for Java classes, assisting 70+ students with AP Computer Science and USACO material.
- Taught beginner Java class for students age 7-11, creating lesson plans, and guiding students in final projects.

Project Experience

Chatbot Development

- Engineered a Python-based Discord bot for remote roleplaying game management, automating game rule calculations previously done manually with dice.
- Self-taught through online documentation, implemented features streamlining battles for groups of friends.

3D Printing

- Authored a user guide facilitating integration of 3D printing support into a 3D software group's workflow.
- Assembled an Original Prusa i3 MK3S+ from a kit, configured Ender 3 printers, producing diverse 3D models.

Skills

Programming: C/C++, Python, Go, Java, Javascript, HTML/CSS
Software: Git, Docker, Perforce, VS Code, macOS, Linux, visual debugger

Activities & Awards

- **Violinist**, University of Michigan Campus Symphony Orchestra, MA All-State Orchestra Jul 2010 - Present
- **Michigan Hackers Core Member,** University of Michigan Jan 2024
- **University Honors and Dean's Honor List 2023, 2024, 2025,** University of Michigan Feb 2025
- **James B. Angel Scholar,** University of Michigan Feb 2025
- **College of Engineering Scholarship Award,** University of Michigan Aug 2024